

ABHISHEK SINGH

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Education

Bachelor of Technology in Computer Science and Engineering
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHAGALPUR

May 2021 - May 2025
CGPA:8.35/10 (Absolute Grading)

Work Experience

Crosslynx

Nov 2024 - Ongoing

Associate Data Scientist

On-Site

- Developed an **unsupervised AI solution** to detect electricity theft at the DTR level using **HDBSCAN clustering**, capable of identifying six distinct theft patterns and outperforming prior methods in **AUC and MAP metrics**.
- Implemented a **RAG-based chatbot** to handle client queries related to the **smart meter installation manual**, **fine-tuned the Nomic embedding model** to improve retrieval accuracy, and applied **advanced memory techniques** leveraging **Agentic AI** to enhance user experience by avoiding repetitive answers and maintaining conversational context.

Technical Skills

Programming Languages: C/C++, Python, SQL

AI/ML Tools: OpenCV, NLTK

Libraries/Frameworks: TensorFlow, PyTorch, LangChain, RAG, LangGraph, FastAPI

Other Skills: Chroma, Faiss, Docker

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, OOPS, DBMS, Generative AI

Publications

- Attention-Guided Encoder-Decoder Design for Efficient RIS Phase Configuration in Time-Varying IoT Networks**
Accepted in *IEEE Transactions on Artificial Intelligence*, 2025 [Paper]
- Federated Learning-Based Fault-Tolerant RIS Reconfiguration for Multi-User IoT Networks**
Accepted in *IEEE Advanced Networks and Telecommunications Systems (ANTS)*, 2025.

Projects

Small Language Model | *Pytorch, Generative AI, Transformer, Docker, Fast API* [Git](#) **July 2024 – August 2024**

- Developed a compact **Small Language Model (SLM)** with **16M parameters** using a **Transformer Decoder** architecture in **PyTorch**, capable of generating coherent English at a smaller scale compared to LLMs.
- Implemented **Key-Value (KV) Caching** to optimize auto-regressive inference by reducing redundant attention computations, significantly improving response latency.
- Containerized** the complete application using **Docker** to enable scalable and portable deployment across environments.

Finetuned-Llava | *Pytorch, Generative AI, Transformer* [Git](#) [HuggingFace](#) **August 2024 – September 2024**

- Fine-tuned Llava using **PEFT** techniques on **Naver Clova IX CORD-v2 dataset** to prevent catastrophic loss and achieve **30% memory reduction**.
- Enabled structured JSON extraction from image receipts for efficient document analysis on **low-resource devices**.

AutoPitch | *Generative AI, Prompt Engineering, LLM* [Git](#) **January 2024 – March 2024**

- Engineered a custom email automation application using Meta's **LLaMA 3.1 model** and **Groq**, resulting in a **30% reduction** in manual client outreach.
- Integrated **LangChain WebLoader** and **ChromaDB** to **automate** job description analysis and semantic link extraction, enhancing email relevance by **25%**.

Achievements

- Received a **Letter of Recommendation** for exceptional research contribution in the field of 6G communication from IIT BHU.
- Qualified twice at the regional level in the International Mathematics Olympiad (IMO).
- Serving as TPC member at **IEEE ANTS & ELEXCOM**, reviewing and evaluating AI research papers to ensure technical quality and relevance.
- Solved 700+ Leetcode problems, including 50 SQL questions, earning the SQL Badge.